

Study Guide for Final Test

Note: This study guide is meant to help you pinpoint specific topics that will be on the upcoming exam. All information covered in lecture and assigned readings (Ch. 27, 28, and 32) is fair game for the test regardless of whether it is mentioned on this study guide.

The final test is cumulative meaning anything we have covered this semester may be on the test. However, there will be a slight bias towards the chapters that have been covered since test 3. The previous study guides will help you to review material covered earlier in the semester. As you review, try to sum up the key points and themes from each chapter. Learning objectives will be helpful as well.

Cortex

Describe general trends of cellular organization in layers of the cerebral cortex

What are some of the broad functions of the temporal lobe?

How is language ability impacted after damage to Wernicke's area?

List the parts of the brain included in the medial temporal lobe system.

Describe the role of each area in declarative memory formation.

Describe how cortical representations of specific objects may be coded within the ventral stream.

What is the difference between sparse coding and population coding?

Which appears to be more plausible for coding specific objects? Why?

What is agnosia?

Describe visual agnosia and prosopagnosia.

What are some of the broad functions of the parietal lobe?

Describe the relationship between the parietal lobe and attention (especially orienting attention)

Describe how certain types of parietal lobe damage help elucidate the relationship.

What are some of the broad functions of the frontal lobe?

What are the different functional subdivisions of prefrontal cortex?

Describe the main purpose of each subdivision.

How do these subdivisions help support executive functions?

Cortical States

What is circadian rhythm?

How does exposure to light influence circadian rhythm?

What part of the brain determines circadian rhythm?

Describe the molecular cycle of this biological clock.

What are the different stages of a sleep cycle?

How can the different sleep stages be distinguished from one another?

What parts of the brain are important for the different stages of sleep and transitions between them?

In what order do we typically progress through these stages?

How does the amount of time spent in each stage change throughout a night of sleep?

How does the amount of time spent in each stage change as we get older?

What are some of the consequences of not getting enough sleep (sleep deprivation)?

Why might sleep be adaptive?

What is the default-mode network?

Which parts of the brain are part of the network?

Describe the relationship between the DMN and the frontoparietal network (voluntary attention)

Describe the clinical relevance of DMN dysfunction.

Emotion

Describe the 3 components that contribute to the overall experience of an emotion.

Which parts of the brain are involved in experiencing emotion?

How might the amygdala play a special role in emotion?

How might the direct and indirect routes to the amygdala influence the experience of emotion?

Describe the relationship between the amygdala and parts of prefrontal cortex as it relates to emotion regulation.

Describe the relationship between the amygdala and the hypothalamus as it relates to the HPA axis.

What symptoms are characteristic of major depressive disorder (MDD)?

What evidence is there that genetics play a role in MDD?

How do the brain structures of individuals diagnosed with MDD differ from people in the general population?

How does brain function differ?

How might these differences relate to symptomology?

How does brain chemistry of individuals diagnosed with MDD differ from people in the general population?

How might these differences relate to symptomology?

What treatments are used for symptoms of MDD?

What does the effectiveness of these treatments tell us about biological factors that may underlie MDD?

What symptoms are characteristic of Bipolar Disorder (BPD)?

What is bipolar cycling?

How do brain structures of individuals diagnosed with BPD differ from people in the general population?

How does brain function differ?

How might these differences relate to symptomology?

What symptoms are characteristic of post-traumatic stress disorder (PTSD)?

How do brain structures of individuals diagnosed with PTSD differ from people in the general population?

How does brain function differ?

How might these differences relate to symptomology?